

Databases

Laboratory Work 1

ER-Diagram of the International Airport Database

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Tool used: PostgreSQL / JetBrains DataGrip

1. System Description Summary

The International Airport Database manages airport operations, including passenger bookings, flights, airlines, boarding, baggage, and security. Based on the system description, ten entities were identified, listed and detailed below with their attributes, data types, and keys.

2. Entities and Attributes

1. airport

Stores information about each airport used in the system, including its location and audit timestamps.

Attribute	Data Type	Key	Description
airport_id	integer	PK	Unique identifier of the airport
airport_name	varchar(100)		Full name of the airport
country	varchar(50)		Country where the airport is located
state	varchar(50)		State/region of the airport
city	varchar(50)		City of the airport
created_at	timestamp		Record creation date/time
updated_at	timestamp		Record last update date/time

2. airline

Stores information about the airline companies operating flights.

Attribute	Data Type	Key	Description
airline_id	integer	PK	Unique identifier of the airline
airline_code	varchar(10)	UK	Unique short code of the airline
name	varchar(100)		Airline company name
country	varchar(50)		Country of the airline
created_at	timestamp		Record creation date/time
updated_at	timestamp		Record last update date/time

3. passenger

Stores passenger profile information associated with bookings.

Attribute	Data Type	Key	Description
passenger_id	integer	PK	Unique identifier of the passenger
first_name	varchar(50)		Passenger's first name
last_name	varchar(50)		Passenger's last name

Attribute	Data Type	Key	Description
gender	varchar(10)		Passenger's gender
date_of_birth	date		Passenger's date of birth
country_of_citizenship	varchar(50)		Passenger's citizenship
country_of_residence	varchar(50)		Passenger's country of residence
passport_number	varchar(20)	UK	Unique passport number
created_at	timestamp		Record creation date/time
updated_at	timestamp		Record last update date/time

4. flight

Stores scheduled and actual flight information, including gates and timings; each flight belongs to one airline and connects two airports.

Attribute	Data Type	Key	Description
flight_id	integer	PK	Unique identifier of the flight
airline_id	integer	FK -> airline.airline_id	Airline operating the flight
departure_airport_id	integer	FK -> airport.airport_id	Departure airport
arrival_airport_id	integer	FK -> airport.airport_id	Arrival airport
departing_gate	varchar(10)		Gate used for departure
arriving_gate	varchar(10)		Gate used for arrival
scheduled_departure_time	timestamp		Planned departure time
scheduled_arrival_time	timestamp		Planned arrival time
actual_departure_time	timestamp		Actual departure time
actual_arrival_time	timestamp		Actual arrival time
created_at	timestamp		Record creation date/time
updated_at	timestamp		Record last update date/time

5. booking

Stores a passenger's booking made for a specific flight.

Attribute	Data Type	Key	Description
booking_id	integer	PK	Unique identifier of the booking
flight_id	integer	FK ->	Booked flight

Attribute	Data Type	Key	Description
		flight.flight_id	
passenger_id	integer	FK -> passenger.passenger_id	Passenger who made the booking
status	varchar(20)		Booking status (e.g. confirmed, cancelled)
booking_platform	varchar(50)		Platform used to make the booking
ticket_price	numeric(10,2)		Price paid for the ticket
created_at	timestamp		Record creation date/time
updated_at	timestamp		Record last update date/time

6. booking_change

Stores the history of changes made to an existing booking.

Attribute	Data Type	Key	Description
change_id	integer	PK	Unique identifier of the change record
booking_id	integer	FK -> booking.booking_id	Booking that was changed
change_description	varchar(255)		Description of the change made
created_at	timestamp		Date/time the change was recorded

7. boarding_pass

Stores the boarding pass issued to a passenger for a booking.

Attribute	Data Type	Key	Description
boarding_pass_id	integer	PK	Unique identifier of the boarding pass
booking_id	integer	FK -> booking.booking_id	Related booking
seat	varchar(10)		Assigned seat number
boarding_time	timestamp		Boarding time
created_at	timestamp		Record creation date/time
updated_at	timestamp		Record last update date/time

8. baggage

Stores baggage registered by a passenger for a booking.

Attribute	Data Type	Key	Description
baggage_id	integer	PK	Unique identifier of the baggage item

Attribute	Data Type	Key	Description
booking_id	integer	FK -> booking.booking_id	Related booking
weight_kg	numeric(5,2)		Weight of the baggage in kilograms
created_at	timestamp		Record creation date/time
updated_at	timestamp		Record last update date/time

9. baggage_checking

Stores the results of checking a registered baggage item at the airport.

Attribute	Data Type	Key	Description
checking_id	integer	PK	Unique identifier of the check record
baggage_id	integer	FK -> baggage.baggage_id	Checked baggage item
passenger_id	integer	FK -> passenger.passenger_id	Passenger who owns the baggage
check_result	varchar(50)		Result of the baggage check
created_at	timestamp		Record creation date/time
updated_at	timestamp		Record last update date/time

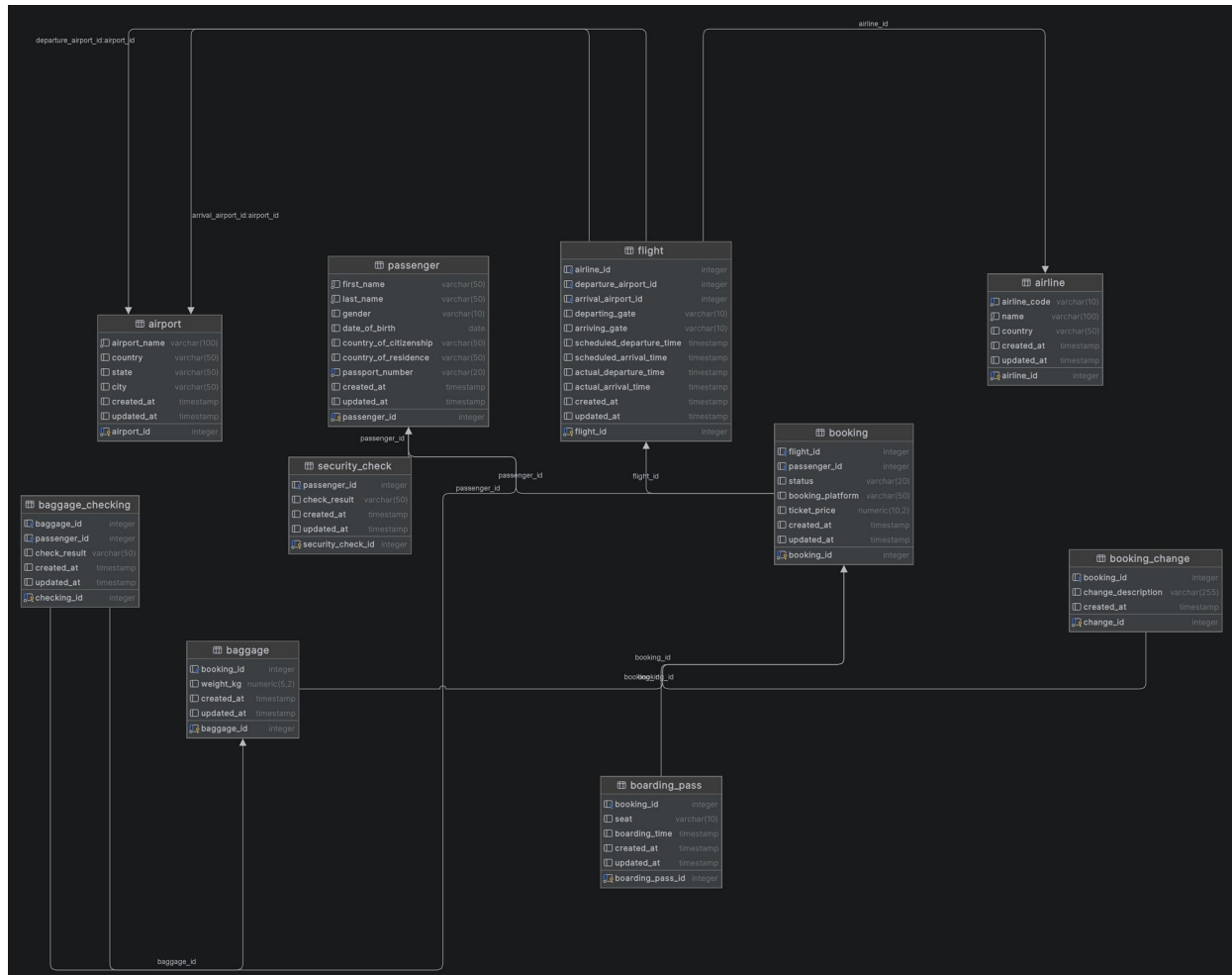
10. security_check

Stores the results of the security check a passenger passes at the airport.

Attribute	Data Type	Key	Description
security_check_id	integer	PK	Unique identifier of the security check
passenger_id	integer	FK -> passenger.passenger_id	Passenger who was checked
check_result	varchar(50)		Result of the security check
created_at	timestamp		Record creation date/time
updated_at	timestamp		Record last update date/time

3. ER Diagram

The diagram below was generated in JetBrains DataGrip from the physical PostgreSQL schema created for this database.



4. Legend

- PK — Primary Key: uniquely identifies each record in the entity.
- FK — Foreign Key: references the primary key of another (parent) entity, implementing the relationship.
- UK — Unique Key: attribute whose value must be unique across all records (not a relationship key).
- Arrow (→): points from the foreign key column in the child (many-side) entity to the primary key column in the parent (one-side) entity.
- 1 : M — One-to-Many relationship between a parent and a child entity.

5. Relationships and Cardinality

All relationships in this schema are One-to-Many, implemented through foreign key constraints. No Many-to-Many relationship exists in this schema, as every linking scenario (e.g. baggage checking) is resolved by direct foreign keys rather than a separate junction table.

Parent Entity (1)	Child Entity (M)	Cardinality	Description
airline	flight	One-to-Many	One airline operates many flights; each flight belongs to exactly one airline.
airport	flight (departure)	One-to-Many	One airport can be the departure point for many flights; each flight has exactly one departure airport.
airport	flight (arrival)	One-to-Many	One airport can be the arrival point for many flights; each flight has exactly one arrival airport.
flight	booking	One-to-Many	One flight can have many bookings; each booking is made for exactly one flight.
passenger	booking	One-to-Many	One passenger can make many bookings; each booking belongs to exactly one passenger.
booking	booking_change	One-to-Many	One booking can have many recorded changes; each change record belongs to one booking.
booking	boarding_pass	One-to-Many	One booking can be associated with boarding pass(es); each boarding pass belongs to one booking.
booking	baggage	One-to-Many	One booking can include several baggage items; each baggage item belongs to one booking.
baggage	baggage_checking	One-to-Many	One baggage item can have one or more check records; each check record refers to one baggage item.
passenger	baggage_checking	One-to-Many	One passenger can own several checked baggage items; each check record refers to one passenger.
passenger	security_check	One-to-Many	One passenger can pass through security multiple times; each security check record belongs to one passenger.